

Session #2-3: Multi-armed Bandits

- (1) Explain how the feedbacks considered in RL is evaluative not instructive, with an example.
- (2) Why do you say, Multi-armed bandit is RL problem in non-associative setting? Explain.
- (3) Provide one applications which can be modelled as Multi-armed bandit? Why do you think this modelling is appropriate?
- (4) If the action value is constant for all the actions, do you need to explore? Why or why not.
- (5) Tabulate approaches we have learned to balance exploration and exploitation with your remark on their application.
- (6) What will be your choice of ϵ , (as used in ϵ -greedy), for each of the following case:
 - (a) Deterministic rewards
 - (b) Reward distribution with lesser variance
 - (c) Reward distributions with larger varianceExplain the reasons for your choice.
- (7) In ϵ -greedy action selection with four actions (a,b,c,d; assume a is greedy action) and $\epsilon=0.4$, what is the probability that greedy action and non-greedy actions are selected.
- (8) Consider the general form of the update rule that updates the estimate of action values:
$$\text{NewEstimate} = \text{OldEstimate} + \text{StepSize} [\text{Target} - \text{OldEstimate}]$$
Comment on how the following choice addresses (i) non-stationarity and (ii) higher reward variance,
 - (a) Step size is $1/n$ for each update, where n is the current timestep
 - (b) Step size is constant $[0 \leq \alpha \leq 1]$ for each update
- (9) Effect of exploration due to setting optimistic initial values is temporary. Why? When would you use this hack?
- (10) Try the following exercises from RL, Sutton & Barto, Second Ed.
 - (a) Exercise 2.2, 2.3
- (11) Try to understand the problem statement in exercise 2.7 (of RL, Sutton & Barto, Second Ed.), while the analysis as required is not required.
- (12) Explain how policy-based methods and value based methods are different?
- (13) Write any two approaches (need not be the only ones discussed in the classes) you can use to maintain the estimate of action values?
- (14) How does UCB differ from ϵ -greedy? Explain.
- (15) Given a problem, what factors will you consider in choosing the right ϵ for ϵ -greedy action selection?

Happy Learning!

Readings Required: Sections 2.1-2.7 of Sutton & Barto, Ed-2